



THELOOK-ECOMMERCE BIG DATA ANALYTICS PROJECT

GP 5

MELIBOEV MUHIDDIN

Overview



Dataset Source: Google BigQuery Public Dataset | [LINK](#)

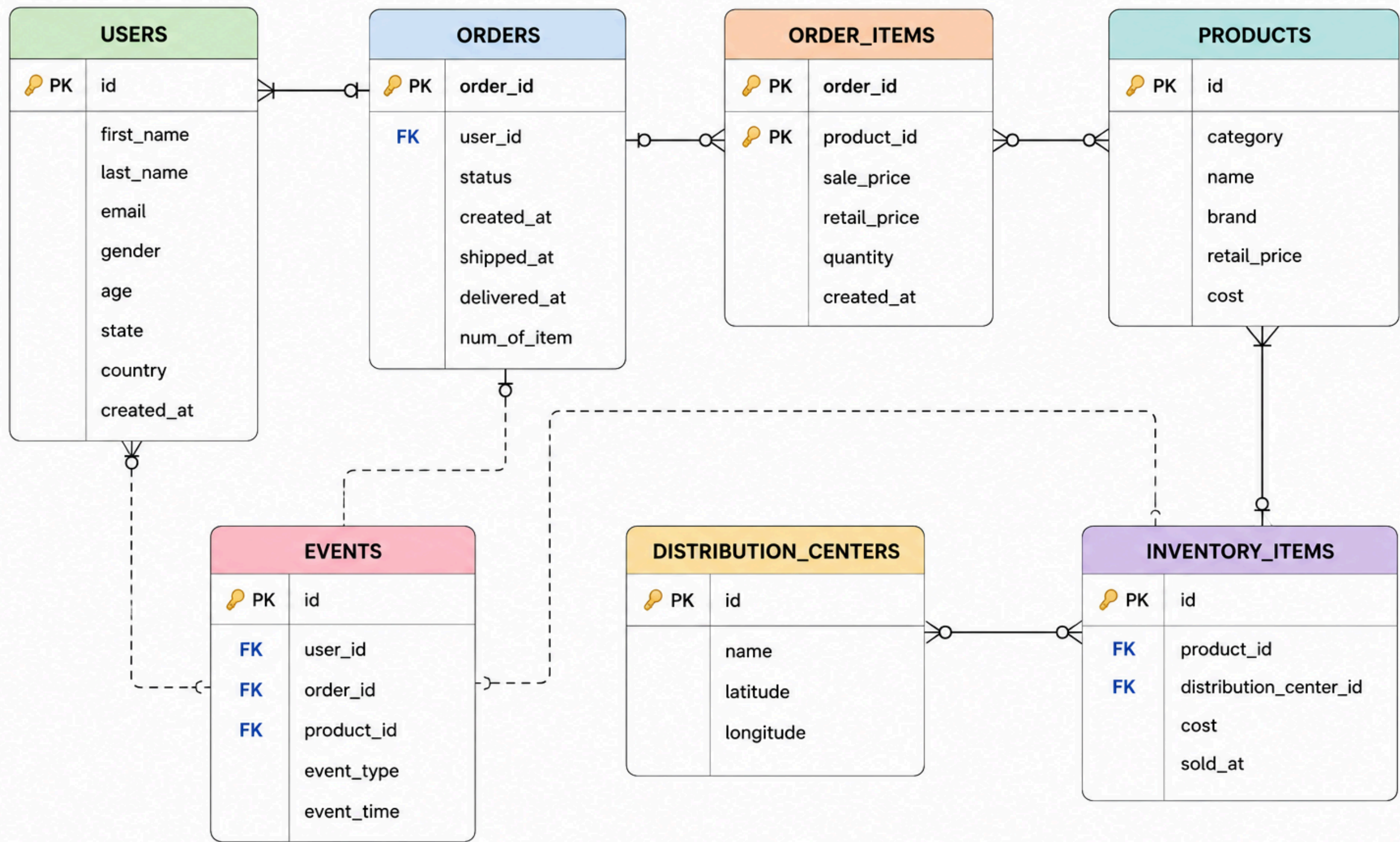
Dataset Name: The_Look_Ecommerce Dataset | [LINK](#)

Business Domain: E-commerce / Retail Analytics

Description:

The dataset represents an online ecommerce business containing customer, order, product, inventory, and behavioral data. It supports customer analytics, sales analysis, operational insights, and business intelligence reporting.

TABLES SCHEMA & ERD OVERVIEW



EXPLORATORY DATA ANALYSIS



Missing values

Duplicates

Outliers

Date range

Important variables





QUERY - TOTAL NUMBER OF USERS

SQL queries 2026-06-06 [Run](#) [Save](#) [Schedule](#) [Share](#)

File Edit Run Tools

```
1 SELECT
2 COUNT(*) AS total_users
3 FROM `bigquery-public-data.thelook_ecommerce.users`;
4
5
6 SELECT
7   p.category,
8   ROUND(SUM(oi.sale_price),2) AS total_revenue
9 FROM `bigquery-public-data.thelook_ecommerce.order_items` oi
10 JOIN `bigquery-public-data.thelook_ecommerce.products` p
11 ON oi.product_id = p.id
12 GROUP BY p.category;
```

✓ This script will process 148.26 MB when run.

Query results [Chat](#) [Save results](#) [Open in](#)

Job information [Results](#) Visualization JSON Execution details Execution graph

Row	total_users
1	100000

TOTAL ORDERS

SQL queries 2026-06-06 [Run](#) [Save](#) [Schedule](#) [Share](#)

File Edit Run Tools

```
1 SELECT
2 COUNT(*) AS total_orders
3 FROM `bigquery-public-data.thelook_ecommerce.orders`;
4
5
6 SELECT
7 COUNT(*) AS total_orders
8 FROM `bigquery-public-data.thelook_ecommerce.orders`;
9
```

✓ Query completed

Query results [Chat](#) [Save results](#) [Open in](#)

Job information [Results](#) Visualization JSON Execution details Execution graph

Row	total_orders
1	125270



QUERY - DATE RANGE ANALYSIS



SQL queries 2026-06-06

Run

Save

Schedule

Share

File

Edit

Run

Tools

9

10

11

12

13

14

SELECT

MIN(

DATE

(

created_at

)

AS

start_date,

MAX(

DATE

(

created_at

)

AS

end_date

FROM

`bigquery-public-data.thelook\_ecommerce.orders`;

✓

This script will process 149.22 MB when run.

Using on-demand processing quota

Query results

Chat

Save results

Open in

Job information

Results

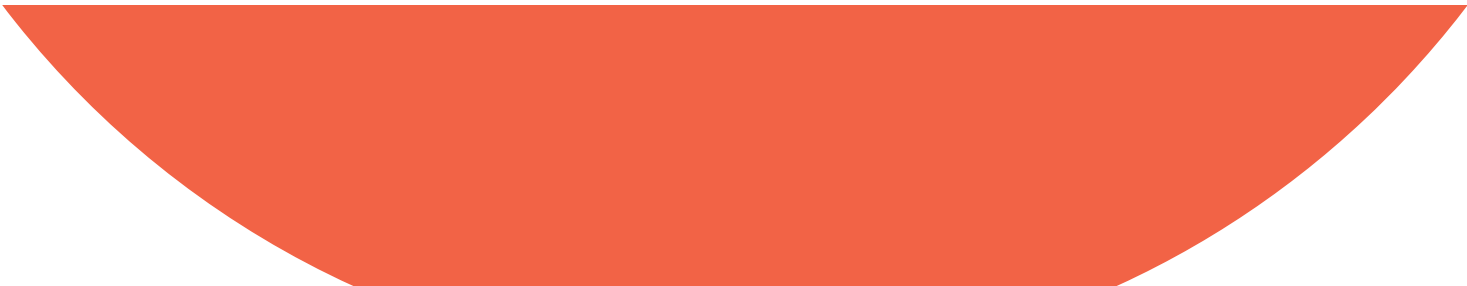
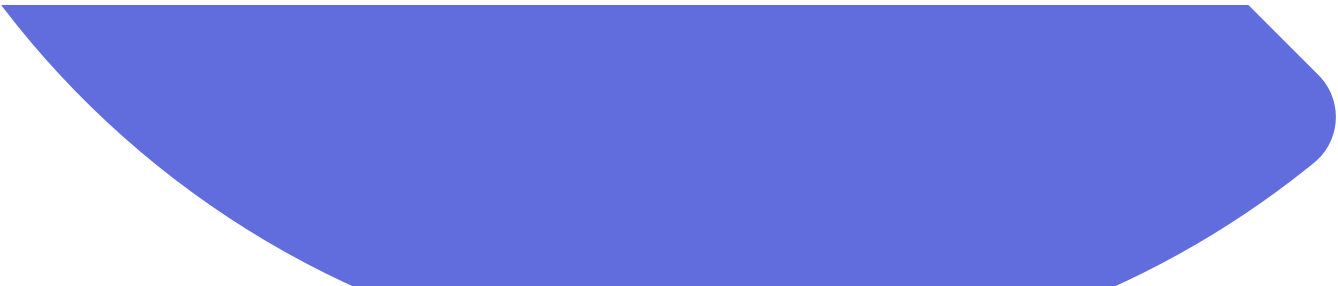
Visualization

JSON

Execution details

Execution graph

Row	start_date	end_date
1	2019-01-10	2026-06-12





QUERY - MISSING VALUES

SQL queries 2026-06-06 Run Save Schedule Share

File Edit Run Tools

```
15 SELECT
16   COUNTIF(first_name IS NULL) AS missing_first_name,
17   COUNTIF(age IS NULL) AS missing_age,
18   COUNTIF(gender IS NULL) AS missing_gender
19 FROM `bigquery-public-data.thelook_ecommerce.users`;
```

✔ This script will process 151.03 MB when run.

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	missing_first_name	missing_age	missing_gender
1	0	0	0

QUERY - DUPLICATE CHECK

SQL queries 2026-06-06 Run Save Schedule Share

File Edit Run Tools

```
20
21 SELECT
22   order_id,
23   COUNT(*) AS duplicate_count
24 FROM `bigquery-public-data.thelook_ecommerce.orders`
25 GROUP BY order_id
```

✔ Query completed

Using on-demand processing quota

Query results Chat Save results Open in

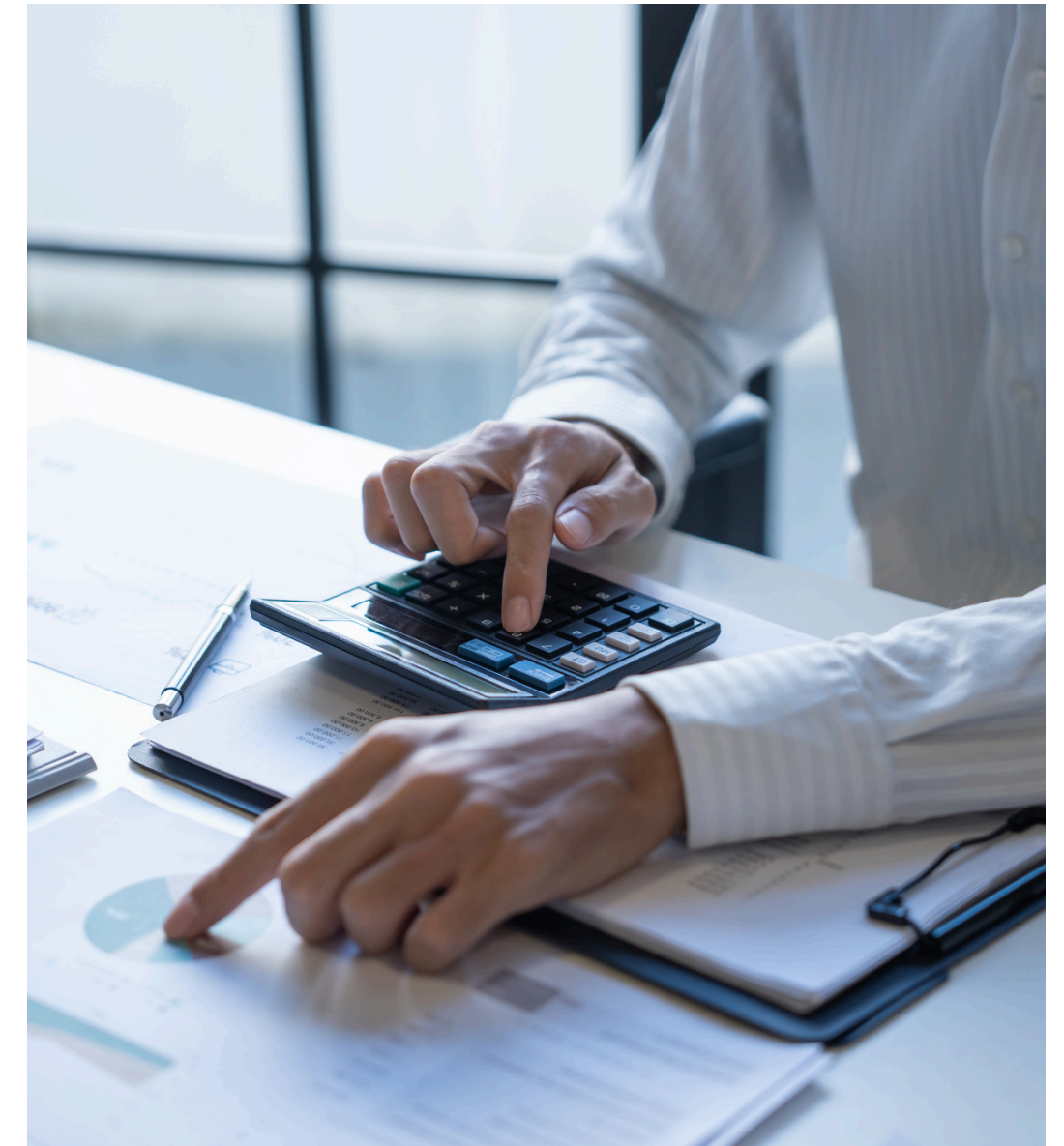
Job information Results Visualization JSON Execution details Execution graph

There is no data to display.

Business Context

TheLook Ecommerce company wants to understand:

- Revenue drivers
- Customer behavior
- Product performance
- Inventory efficiency
- Customer retention
- Purchase journey optimization



BUSINESS QUESTIONS



Business Questions:

- Which product categories generate the highest revenue?
- What are the monthly and seasonal sales trends?
- Which customer groups purchase most frequently?
- Which products have high inventory but low sales?
- Which distribution centers perform most efficiently?
- How does customer behavior influence purchasing activity?

Analytics Framework



Executive Dashboard

TheLook_Ecommerce

Executive Overview Dashboard

Category
All

Gender
All

Date Range
Jan 1, 2019 - Jan 7, 2026

\$2.66M

Total Revenue

31.1K

Total Orders

27.4K

Total Customers

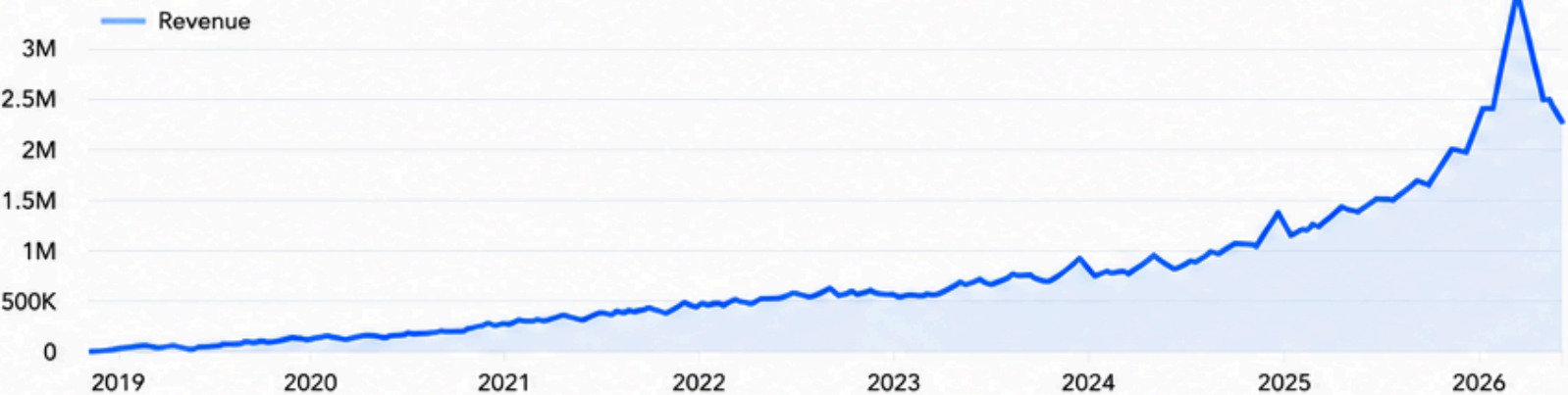
\$85.53

Average Order Value (AOV)

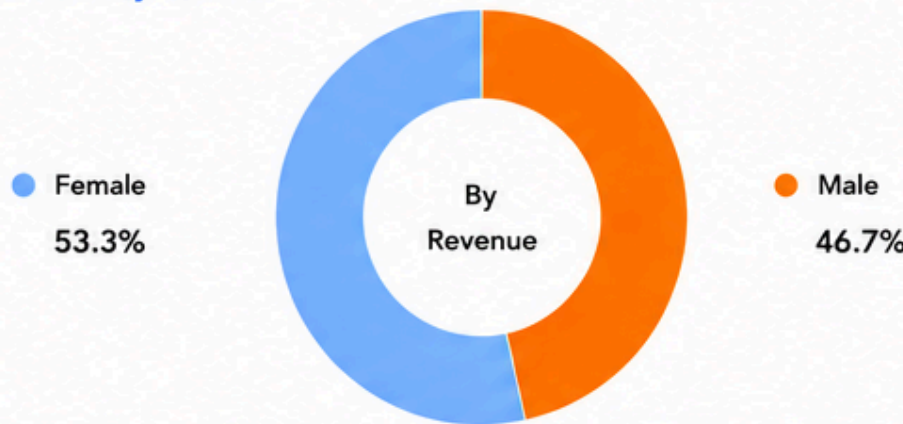
12.39%

Repeat Customer Rate

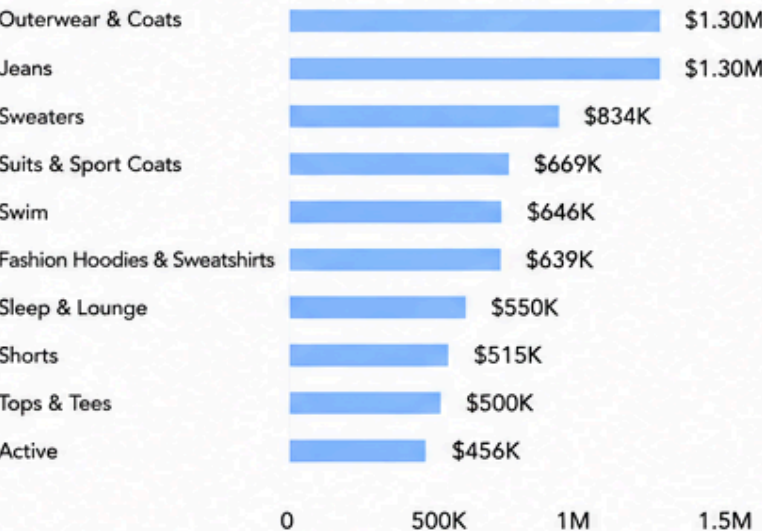
Revenue Trend Over Time



Revenue by Gender



Revenue by Product Category

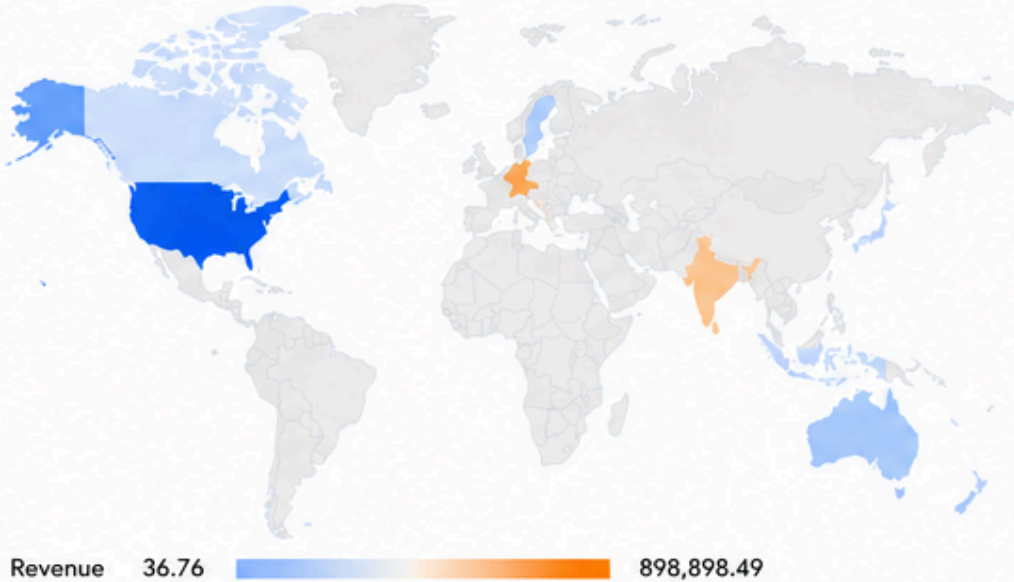


Top 10 Products by Revenue

#	Product Name	Revenue	Units Sold
1	adidas Women's adiFIT Slim Pant	\$3,612	4
2	The North Face Apex Bionic Soft Shell Jacket - Men's	\$3,612	4
3	Quiksilver Men's Rockefeller Walkshort	\$3,612	4
4	Canada Goose Women's Solaris	\$3,475	5
5	Nobis Yatesy Parka	\$2,850	3
6	Jordan Low Quarter Sock Style # 427411	\$2,709	3
7	Patagonia Men's Better Sweater	\$2,662	3
8	Levi's Men's 511 Slim Fit Jean	\$2,166	2
9	Ralph Lauren Men's Crewneck Tee	\$2,035	2
10	Nike Men's Dri-FIT Short Sleeve	\$1,883	2

1 - 10 / 10

Revenue by Country



Key Insights

- Total revenue reached \$2.66M from 31.1K orders, showing consistent growth over time.
- Outerwear & Coats and Jeans are the top revenue generating categories.
- Female customers contribute 53.3% of total revenue, slightly more than males (46.7%).
- Repeat customer rate is low at 12.39%, indicating a strong retention opportunity.
- Focus on customer engagement and loyalty programs to drive repeat purchases.



Q1 What are the monthly and seasonal sales trends?

SQL queries 2026-06-06

Run

Save query

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Schedule

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```
10 SELECT
11   EXTRACT(MONTH FROM created_at) AS month,
12   ROUND(SUM(sale_price),2) AS revenue
13 FROM `bigquery-public-data.thelook_ecommerce.order_items`
14 GROUP BY month
15 ORDER BY month;
16
```

Query completed

Query results

ChatSave resultsOpen in

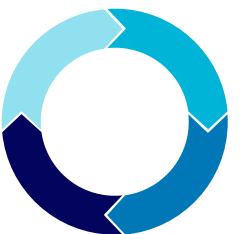
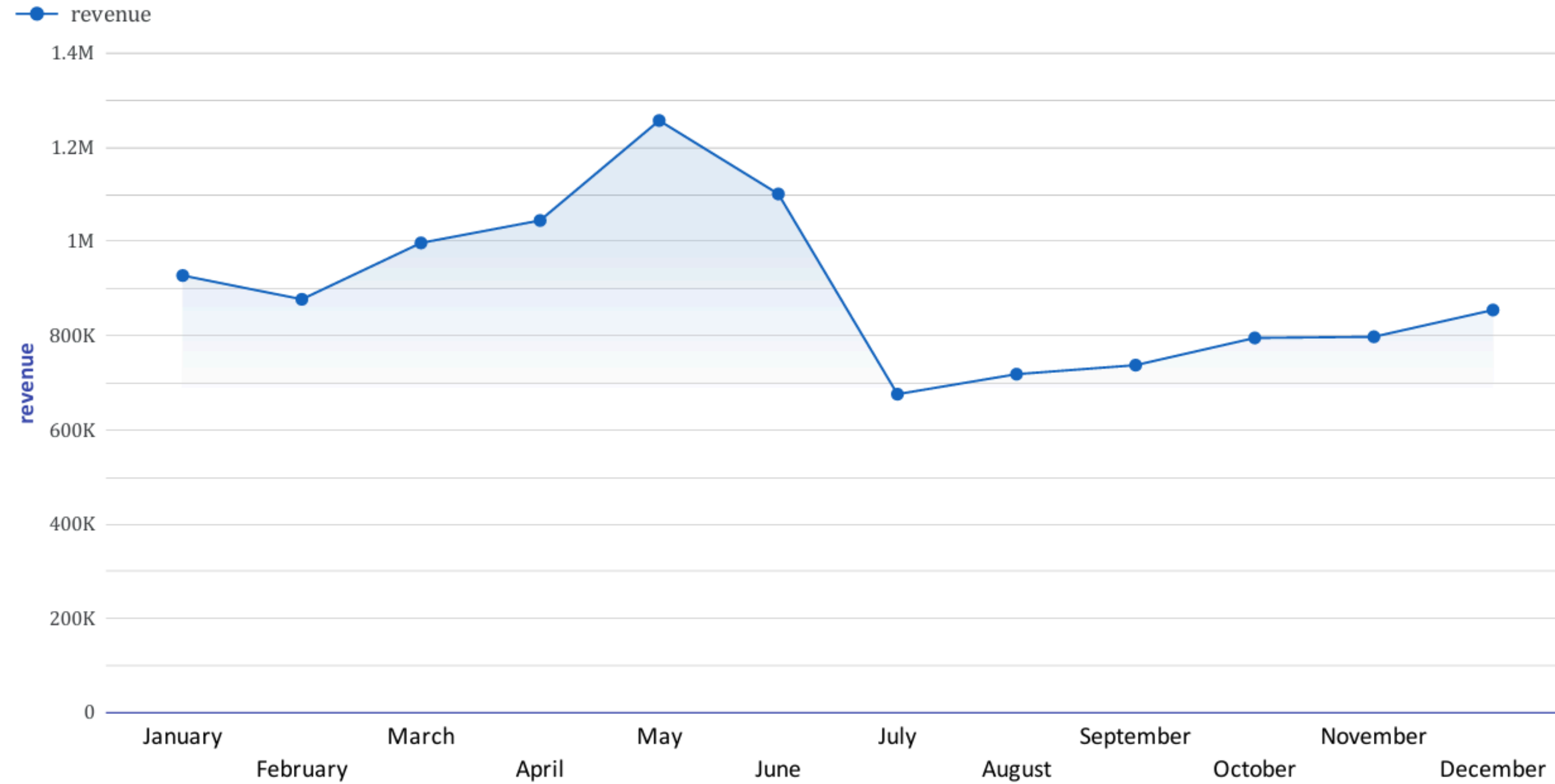
Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	month	revenue
1	1	926535.93
2	2	876070.42
3	3	995687.5
4	4	1043359.34
5	5	1255367.37
6	6	1099624.16
7	7	674735.19



Visualization

revenue trend by month



BUSINESS INSIGHTS for Q1



The analysis shows that revenue fluctuates throughout the year, indicating seasonal variations in customer purchasing behavior. Revenue increased steadily during the first half of the year and reached its highest level in May, while lower sales were observed during the summer months. These trends suggest that customer demand is not evenly distributed across the year and is influenced by seasonal factors.



Recommendation: Management should align inventory planning, staffing, and marketing activities with seasonal demand patterns. Additional promotional campaigns should be implemented during lower-performing periods to stimulate sales, while sufficient inventory should be maintained before peak sales months to maximize revenue opportunities.



Q2

Which product categories generate the highest revenue?

Untitled query

Run

Save

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Schedule

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More

```
1 SELECT
2   p.category,
3   ROUND(SUM(oi.sale_price),2) AS total_revenue
4 FROM `bigquery-public-data.thelook_ecommerce.order_items` oi
5 JOIN `bigquery-public-data.thelook_ecommerce.products` p
6 ON oi.product_id = p.id
7 GROUP BY p.category
8 ORDER BY total_revenue DESC;
```

Query completed

Query results

Chat

Save results

Open in

Job information

Results

Visualization

JSON

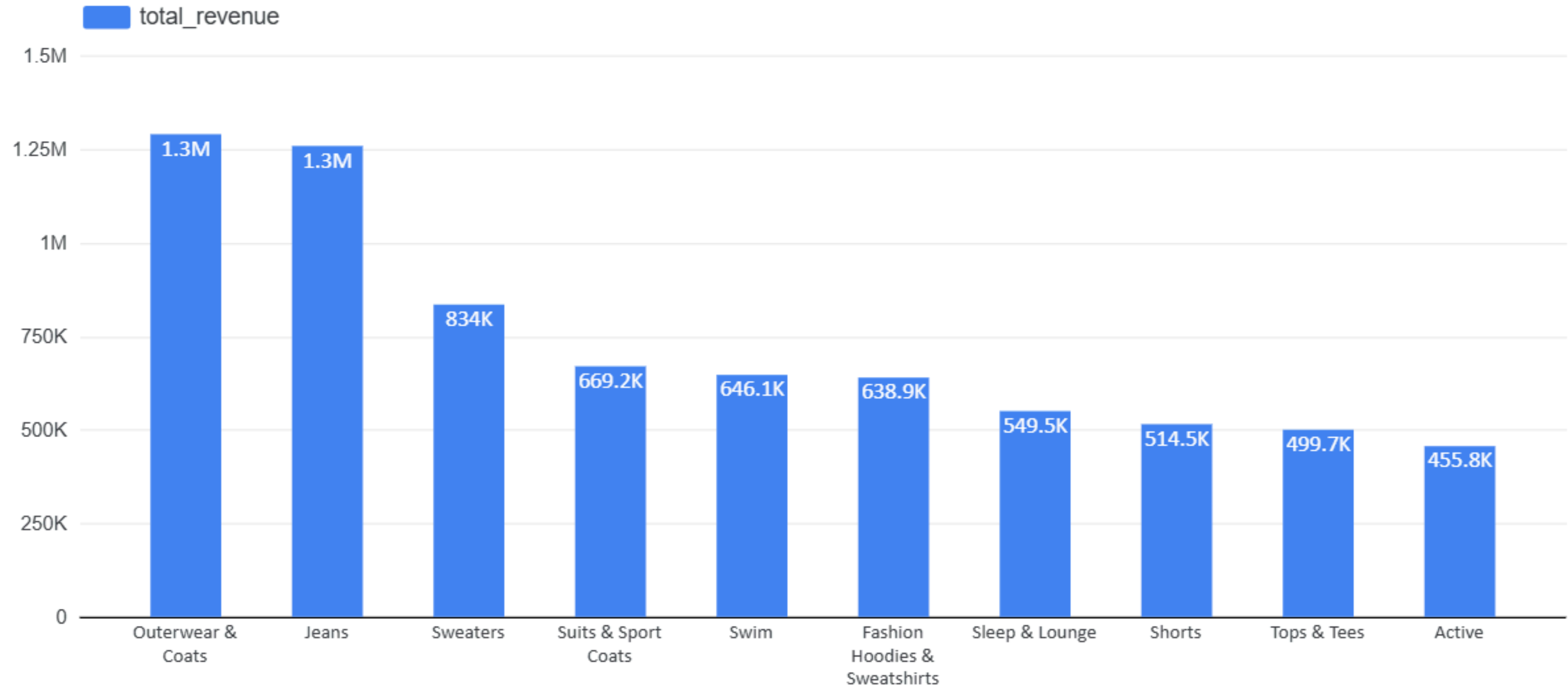
Execution details

Execution graph

Row	category	total_revenue
1	Outerwear & Coats	1320862.79
2	Jeans	1257690.9
3	Sweaters	831844.57
4	Suits & Sport Coats	667665.95
5	Swim	642384.71
6	Fashion Hoodies & Sweatshirts	632990.66
7	Sleep & Lounge	542098.4



Revenue by Product Category



BUSINESS INSIGHTS for Q2



- Outerwear & Coats generated the highest revenue (\$1.32M), followed by Jeans (\$1.26M) and Sweaters (\$0.83M).
- These categories are the primary contributors to overall sales performance.

The results suggest that customers show stronger purchasing demand for apparel categories such as **Outerwear & Coats** and **Jeans**.

Recommendation: Management should prioritize inventory planning, marketing campaigns, and product assortment expansion for top-performing categories to maximize future revenue growth.





Q3

Which customer segments generate the highest revenue?

SQL queries 2026-06-06

Run

Save query

Download

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Schedule

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More

```
10 SELECT
11   EXTRACT(MONTH FROM created_at) AS month,
12   ROUND(SUM(sale_price),2) AS revenue
13 FROM `bigquery-public-data.thelook_ecommerce.order_items`
14 GROUP BY month
15 ORDER BY month;
16
```

Query completed

Query results

Chat

Save results

Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	month	revenue
1	1	926535.93
2	2	876070.42
3	3	995687.5
4	4	1043359.34
5	5	1255367.37
6	6	1099624.16
7	7	674735.19

Dashboard for Q3



Key Insight

Male customers generate a slightly higher share of revenue (53.2%) than female customers (46.8%). Additionally, customers aged 55+ contribute the highest revenue, making them the most valuable customer segment.



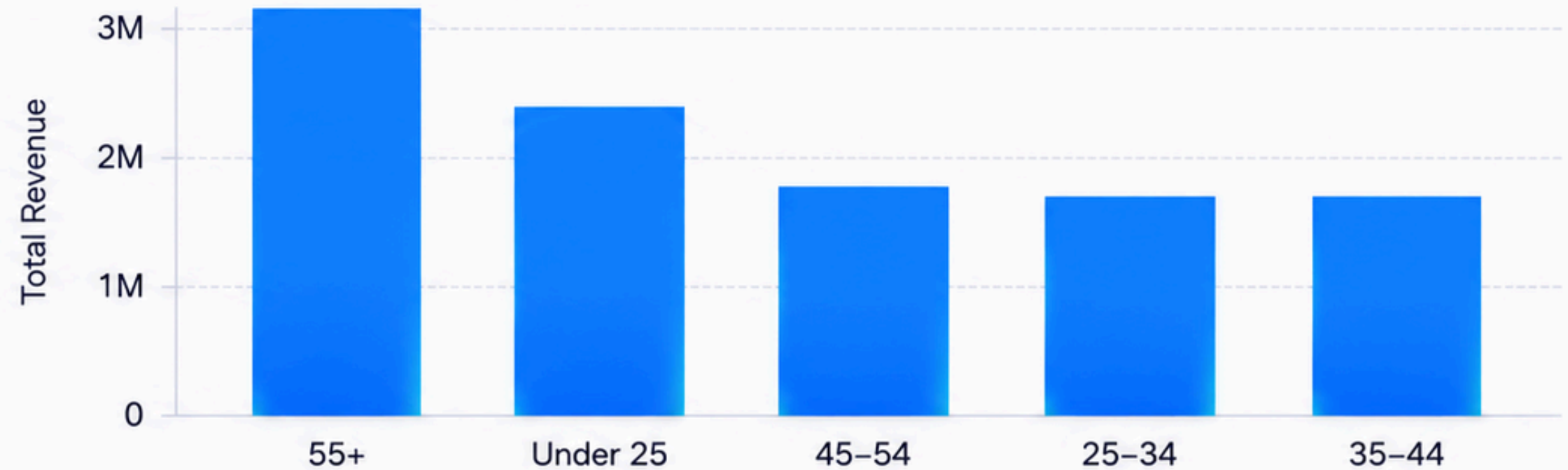
Recommendation

Focus marketing efforts on high-value customer segments, particularly customers aged 55+, while developing targeted campaigns to increase engagement among younger customers.

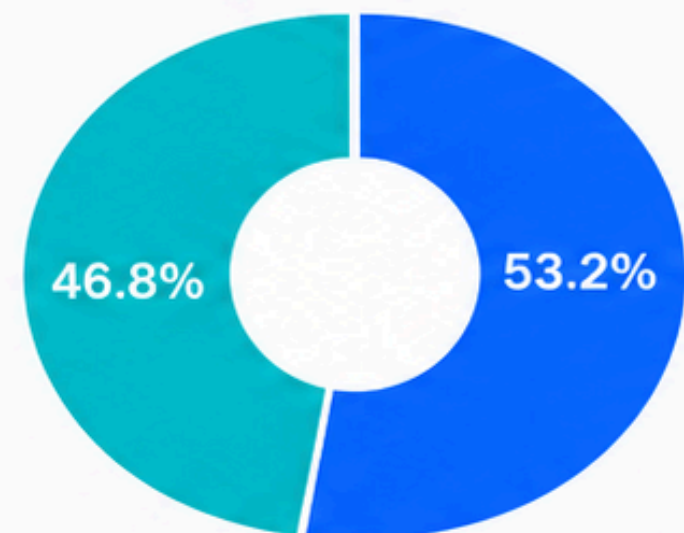


Revenue by Age Group

■ Total Revenue



Revenue Share by Gender



Male (M)	53.2%
Female (F)	46.8%
Total	100%



Q4

Which products have high inventory levels but low sales performance?

SQL queries 2026-06-06

RunSave queryDownloadShareScheduleOpen inMore

```
56 SELECT
57   p.name AS product_name,
58   COUNT(DISTINCT ii.id) AS inventory_count,
59   COUNT(DISTINCT oi.order_id) AS sales_count
60 FROM `bigquery-public-data.thelook_ecommerce.products` p
61 LEFT JOIN `bigquery-public-data.thelook_ecommerce.inventory_items` ii
62   ON p.id = ii.product_id
63 LEFT JOIN `bigquery-public-data.thelook_ecommerce.order_items` oi
64   ON p.id = oi.product_id
65 GROUP BY product_name
66 HAVING inventory_count > 50
```

Query completed

Using on-demand processing quota

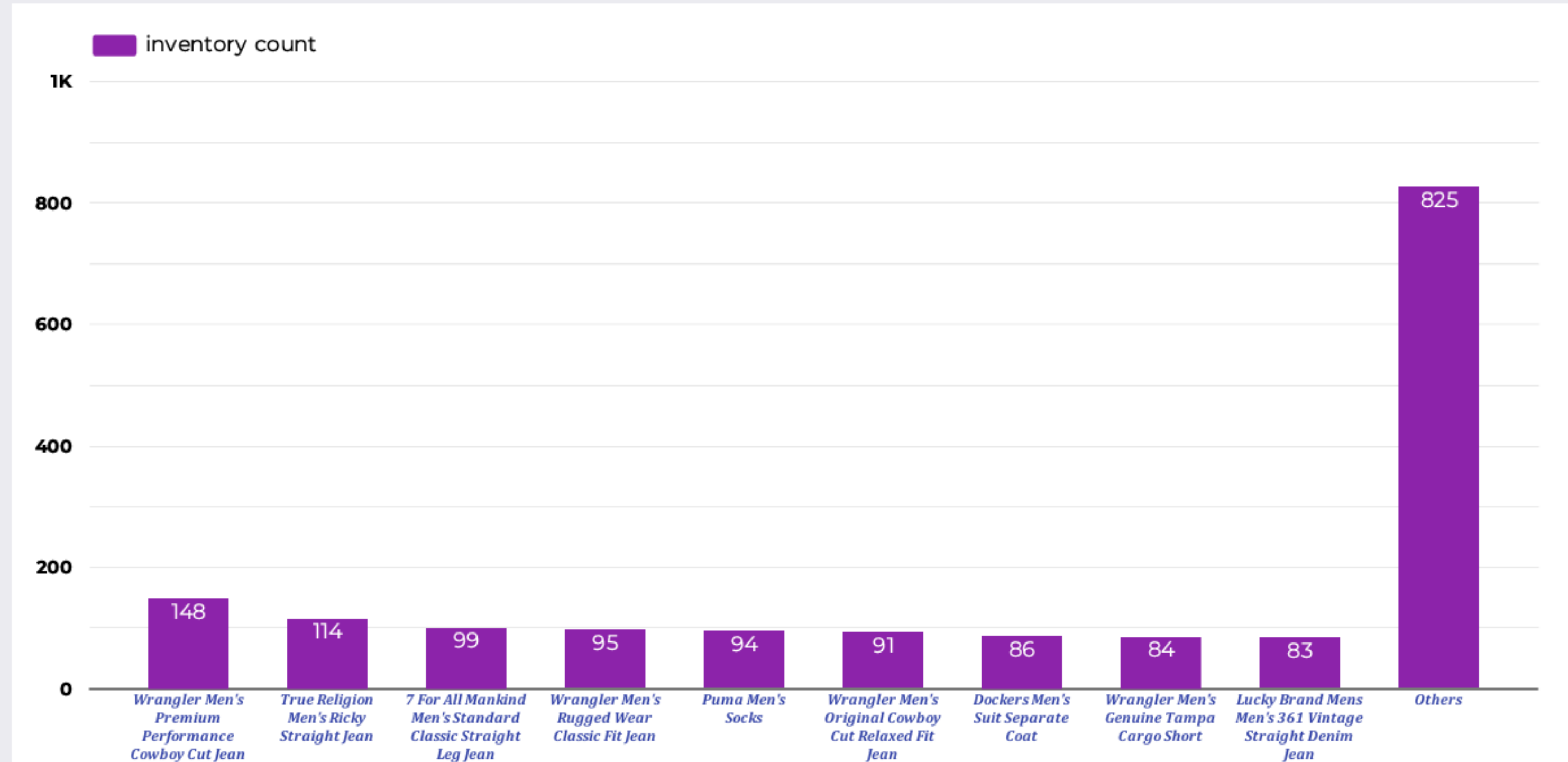
Query results

ChatSave resultsOpen in

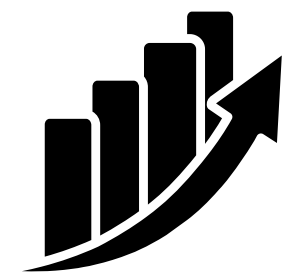
Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	product_name	inventory_count	sales_count
1	Wrangler Men's Premium Perfor...	148	54
2	True Religion Men's Ricky Straig...	114	42
3	7 For All Mankind Men's Standar...	99	38
4	Wrangler Men's Rugged Wear Cl...	95	38

High Inventory, Low Sales Products



BUSINESS INSIGHTS for Q4



Several products maintain high inventory levels despite relatively low sales performance. Products such as **Wrangler Men's Premium Performance** and True Religion **Men's Ricky Straight Jeans** have substantial stock levels but generate comparatively fewer sales, indicating potential overstocking and inefficient inventory utilization.



Recommendation: The company should review inventory planning for slow-moving products and consider promotional campaigns, discounts, or bundle offers to reduce excess stock. Optimizing inventory levels can help lower storage costs and improve overall inventory efficiency.



Q5

Which distribution centers perform most efficiently?

SQL queries 2026-06-06

Run

Save

Schedule

Share

FileEditRunTools

```
185 SELECT
186   dc.name AS distribution_center,
187   COUNT(DISTINCT oi.order_id) AS total_orders,
188   ROUND(SUM(oi.sale_price), 2) AS total_revenue,
189   ROUND(
```

Query completed

Using on-demand processing quota

Query results

Chat

Save results

Open in

Job information

Results

Visualization

JSON

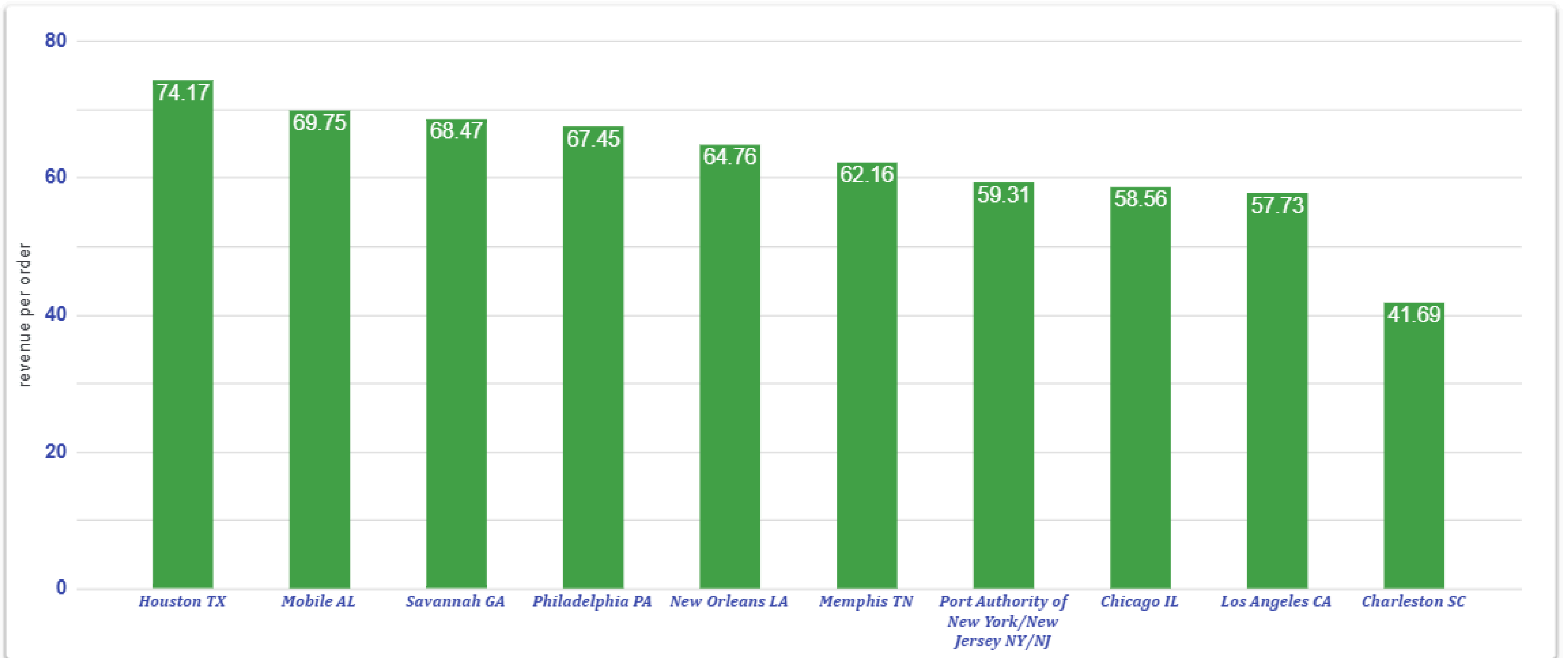
Execution details

Execution graph

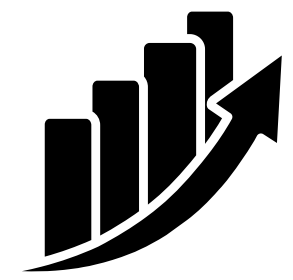
Row	distribution_center	total_orders	total_revenue	revenue_per_order
1	Houston TX	21673	1607515.63	74.17
2	Mobile AL	17483	1219418.61	69.75
3	Savannah GA	11635	796655.44	68.47
4	Philadelphia PA	16226	1094439.61	67.45
5	New Orleans LA	12457	806696.62	64.76
6	Memphis TN	23059	1433420.45	62.16

Vizualization for Q5

revenue per order by distribution center



BUSINESS INSIGHTS for Q5



- Houston, TX achieved the highest operational efficiency with a Revenue per Order of **\$74.17**, outperforming all other distribution centers.
- Mobile, AL and Savannah, GA also demonstrated strong performance, generating over **\$68** revenue per order.
- Memphis, TN processed the highest order volume (23,059 orders) but generated a lower Revenue per Order (**\$62.16**) compared to top-performing centers.
- Significant performance differences exist across distribution centers, indicating opportunities for operational optimization and resource allocation.



Recommendation:

- Prioritize investment and operational best practices from top-performing centers such as Houston, TX and Mobile, AL.
- Investigate factors contributing to lower efficiency in high-volume centers to improve profitability per order.
- Use Revenue per Order as a key performance metric when evaluating distribution center effectiveness and future expansion decisions.
- Reallocate inventory and fulfillment resources toward locations that consistently generate higher revenue per order

Retention Analytics

Understand customer retention behavior and identify churn opportunities

Last Updated
May 11, 2026

Total Customers

80,011

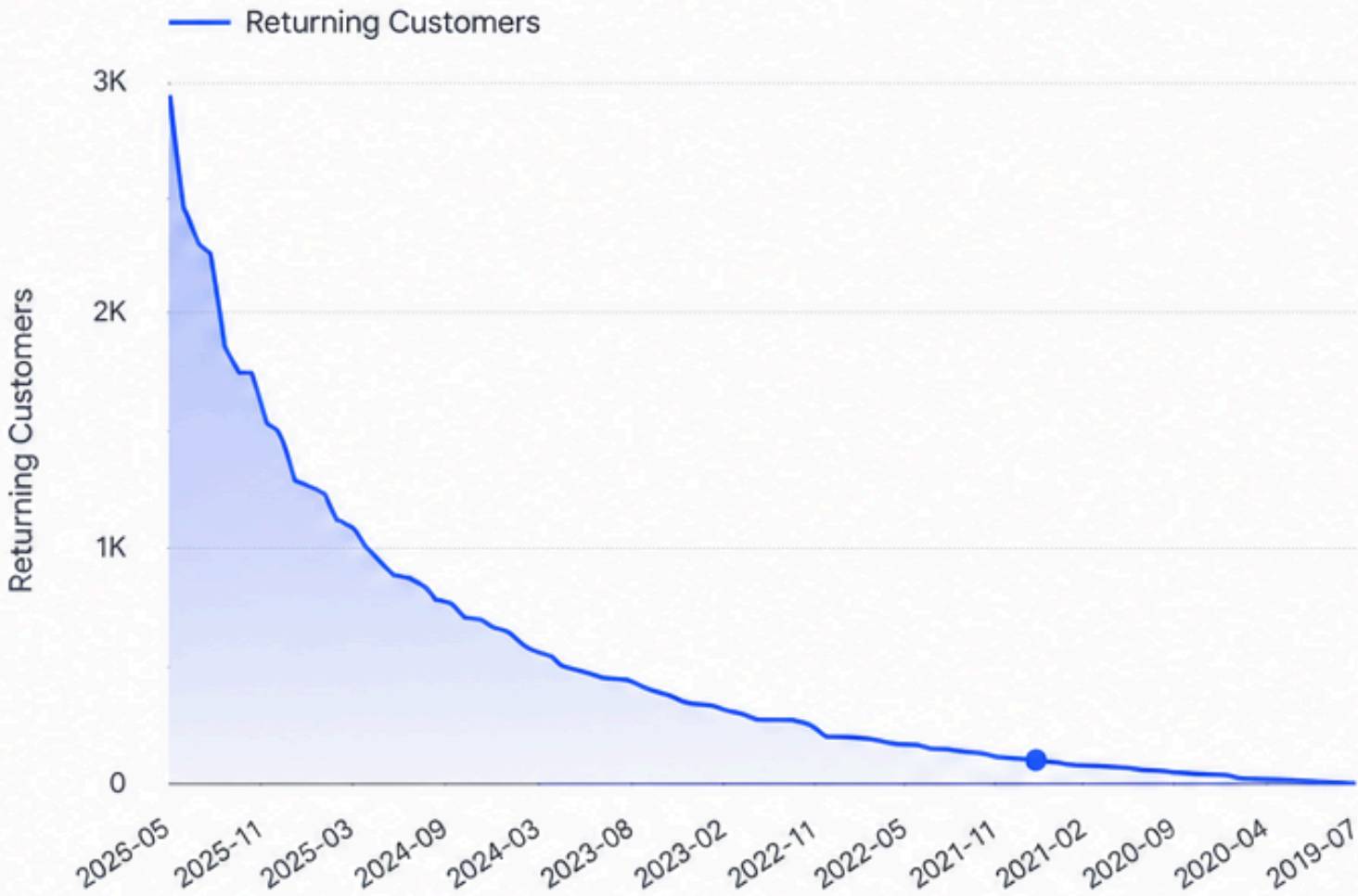
Returning Customers

30,183

Churned Customers

63,244

Returning Customers Over Time



Insight: Returning customers drop significantly after the first few months. Focus on retention strategies to re-engage inactive customers.

Cohort Retention Analysis

Number of users retained in each cohort

Cohort Month		Retained Users ▾
1.	May 1, 2026	3,620
2.	Jun 1, 2026	3,486
3.	Apr 1, 2026	3,187
4.	Mar 1, 2026	3,153
5.	Dec 1, 2025	2,823
6.	Jan 1, 2026	2,762
7.	Nov 1, 2025	2,701
8.	Feb 1, 2026	2,648
9.	Oct 1, 2025	2,566

Funnel Analytics & Conversion Optimization

Understand user behavior across the purchase journey and identify optimization opportunities.

May 1, 2025 - May 11, 2026



Total Product Views

844,688

Total product page views



Cart Events

594,206

Total add to cart events



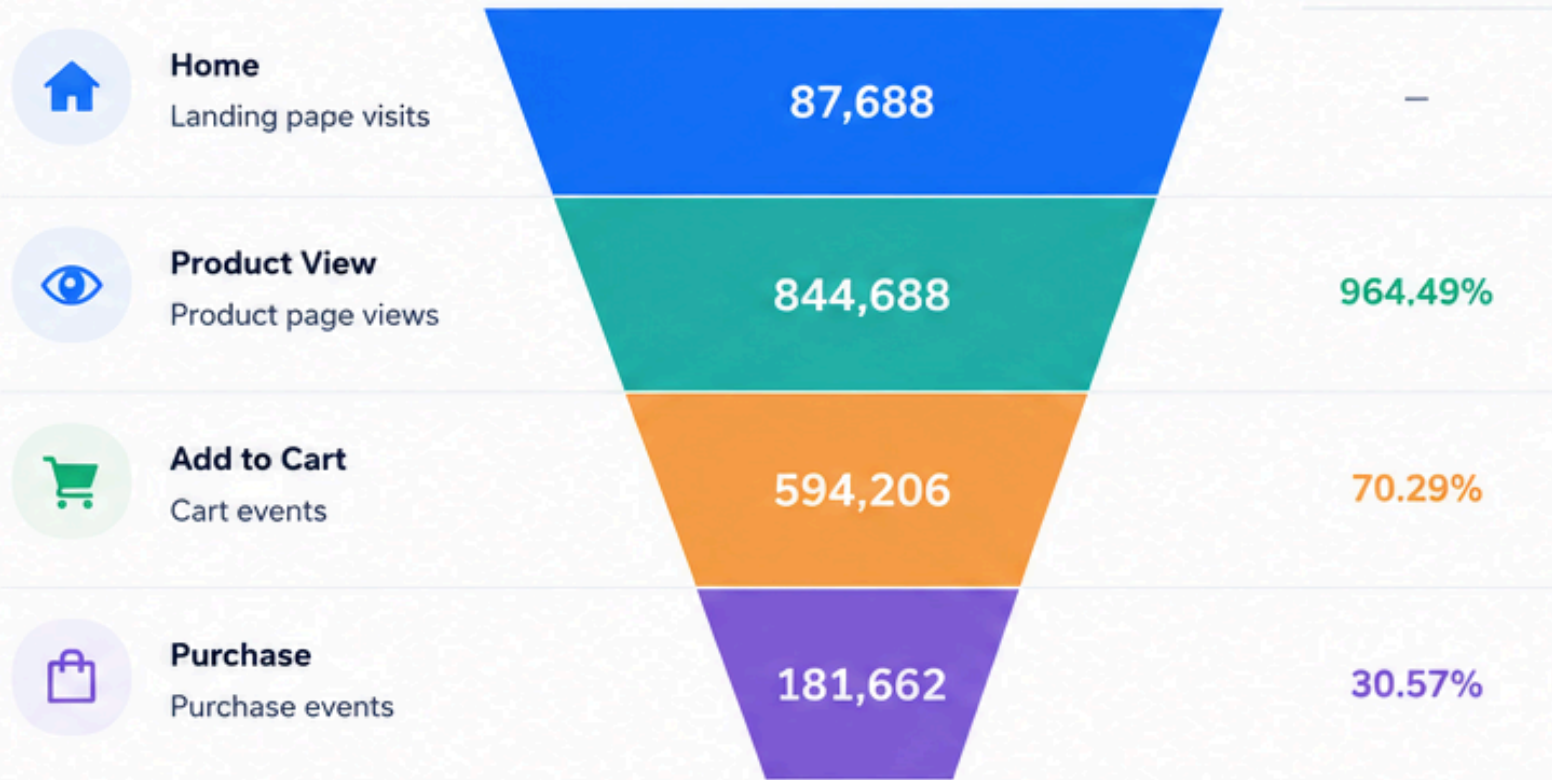
Purchase Events

181,662

Total purchase events

Funnel Overview

Event volume across the user journey



Numbers represent total events for each stage.

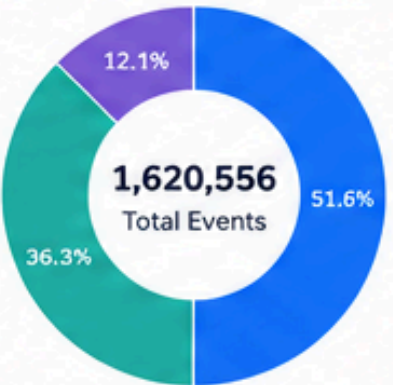
Event Volume by Funnel Stage

Total events by each stage



Purchase Share Across User Journey

Distribution of key events from product view to purchase



Product View	844,688	(51.6%)
Add to Cart	594,206	(36.3%)
Purchase	181,662	(12.1%)
Total	1,620,556	(100%)

Key Insights



High Drop-off After Cart

Only 30.57% of users who add items to cart complete the purchase.



Large Interest, Low Conversion

Product page views are 4.65x higher than purchase events.



Major Optimization Opportunity

Focusing on checkout experience and cart incentives can drive more purchases.

Technical tools:

- SQL(BigQuery)
- Looker studio
- Google Cloud Platform(GCP)

Skills:

- Data Analysis
- Data Cleaning
- Data Visualization
- Dashboard Development
- KPI reporting
- Business Insights





THANK YOU